6 b output.sci (C:\Users\LENOVO\Documents\6 b output.sci) - SciNotes

*6 b output.sci

```
2
3  // Define parameters
4  N = 20; // Number of samples
5  x_n = [ones(1,5), zeros(1,N-5)]; // Rectangular pulse sequence with width 5
6  delay = 3; // Delay of 3 samples
7  y_n = [zeros(1,delay), x_n(1:$-delay)]; // Delayed version of x[n]
8  // Compute the cross-correlation
9  R_xy = correl(x_n, y_n);
10 // Create a lag vector for plotting
11 lags = -N+1:N-1;
12 // Plot the first sequence
13 clf; // Clear the current figure
14 subplot(3,1,1);
15 bar(0:N-1, x_n, 'b');
16 xlabel('n (Discrete Time Index)');
17 ylabel('Amplitude');
18 title('Sequence 1: Rectangular Pulse Sequence');
19 // Plot the second sequence
20 subplot(3,1,2);
21 bar(0:N-1, y_n, 'g');
22 xlabel('n (Discrete Time Index)');
23 ylabel('Amplitude');
24 title('Sequence 2: Delayed Rectangular Pulse Sequence');
25 // Plot the cross-correlation sequence
26 subplot(3,1,3);
27 bar(lags, R_xy, 'r');
28 xlabel('Lag');
29 ylabel('Cross-Correlation');
30 title('Cross-Correlation between Sequence 1 and Sequence 2');
31
32
```

